END-TO-END HEALTHCARE DATA PIPELINE WITH INCREMENTAL PROCESSING WITH ADF AND DATABRICKS

BY :

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OBJECTIVE

The objective of this pipeline is to process data from multiple source systems in a reliable and controlled manner, ensuring that only new data is handled while maintaining accuracy and consistency. It is designed to run smoothly even when no new data is available and to keep a clear record of all activities for tracking and auditing purposes.

This pipeline ingests data from various datasets, processes it through structured layers, and ensures it remains accurate, consistent, and ready for downstream use and analytics.

KEY CAPABILITIES :

- Process only new or unprocessed data to avoid duplication
- Handle no-data scenarios gracefully without causing pipeline failure
- Maintain data integrity through controlled and consistent transformations
- Ensure proper structuring and organization of data across layers
- Log all activities for tracking, monitoring, and auditing purposes
- Support rerunnable and idempotent execution for production reliability
- Enable easy troubleshooting and visibility into pipeline execution
- Ensure scalability to handle increasing data volume over time

SCOPE

This pipeline covers ingestion and processing of multiple datasets from source systems into the data platform. It includes movement of data across raw, bronze, and silver layers, along with validation, transformation, and audit logging.

The pipeline handles structured data files and ensures only new or unprocessed data is moved forward. It supports end-to-end processing from ingestion to cleaned and usable datasets for analytics.

Key Scope Points

- Covers multiple datasets such as providers, visits, claims, and billings
- Includes data ingestion from source to raw storage
- Processes data through bronze and silver layers
- Applies data validation and transformation rules
- Maintains audit logs for each pipeline execution
- Handles incremental data processing
- Supports rerunnable execution without duplication

DATA FLOW

The pipeline ingests data from on-premise systems, cloud sources, and repositories into the raw layer in ADLS using batch ingestion, storing files as received without modification to preserve original data.

Data is then moved to the bronze layer, where it is standardized into a consistent format and organized for processing. Incremental logic is applied to ensure that only new or unprocessed data is handled in each run.

From the bronze layer, data is validated, cleaned, and transformed before being loaded into the silver layer. In this layer, data is structured, deduplicated, and prepared for downstream analytics and reporting.

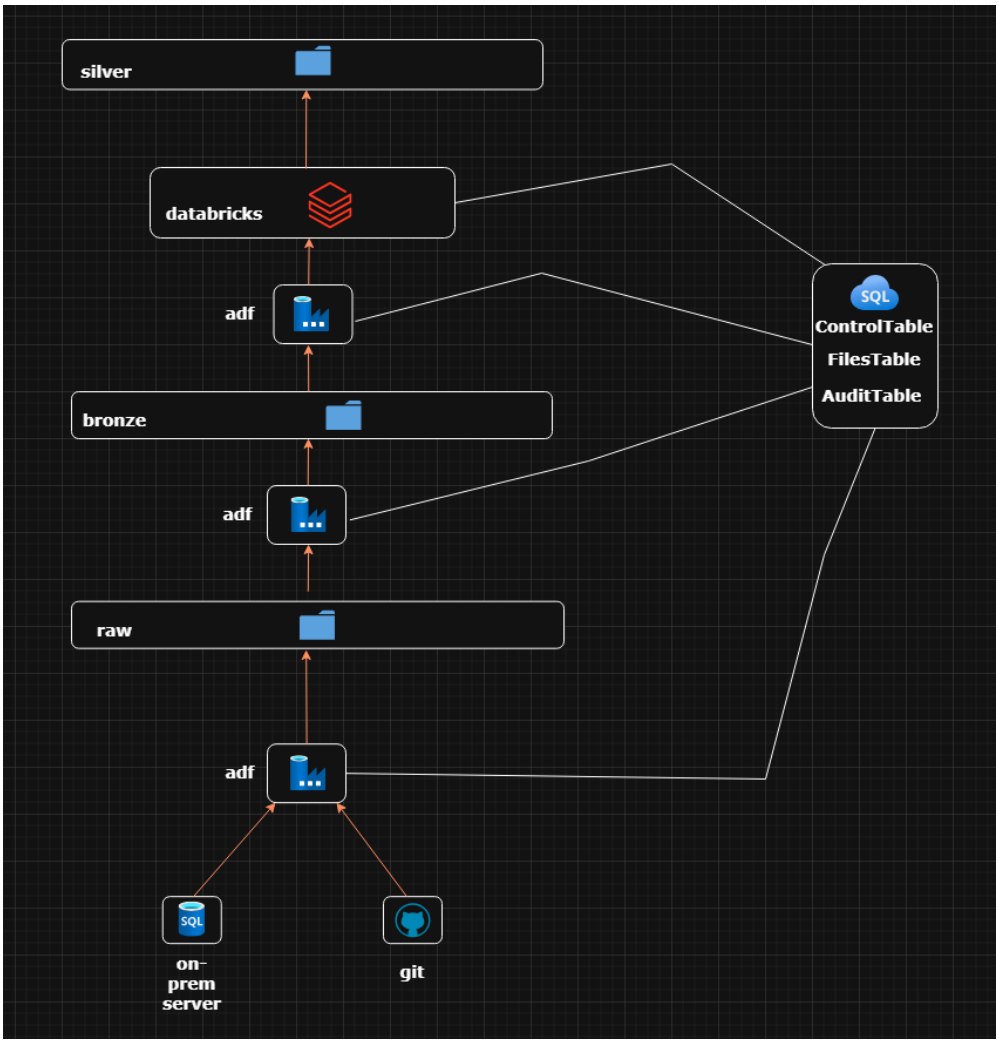
The pipeline maintains audit logs at each stage to track execution status, data movement, and failures. It is designed to be rerunnable and handles no-data scenarios gracefully without causing pipeline failure, ensuring reliability in production environments.

ARCHITECTURE

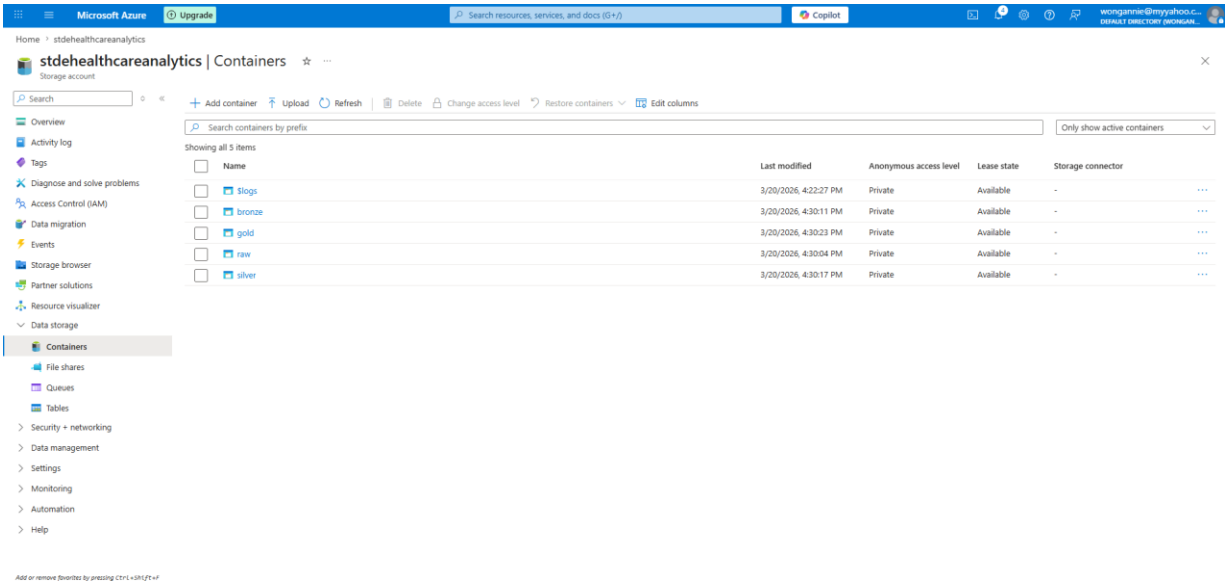
The pipeline is built using Azure Data Factory for orchestration, Azure Data Lake Storage for data storage, and Databricks for data processing and transformation. Data is ingested from on-premise SQL Server, cloud sources, and repositories into ADLS. Data flows through raw, bronze, and silver layers to ensure proper organization, scalability, and performance.

ADF pipelines manage end-to-end orchestration, including batch ingestion, validation, transformation, and audit logging. Databricks notebooks handle data cleaning, deduplication, and business logic before loading into the silver layer.

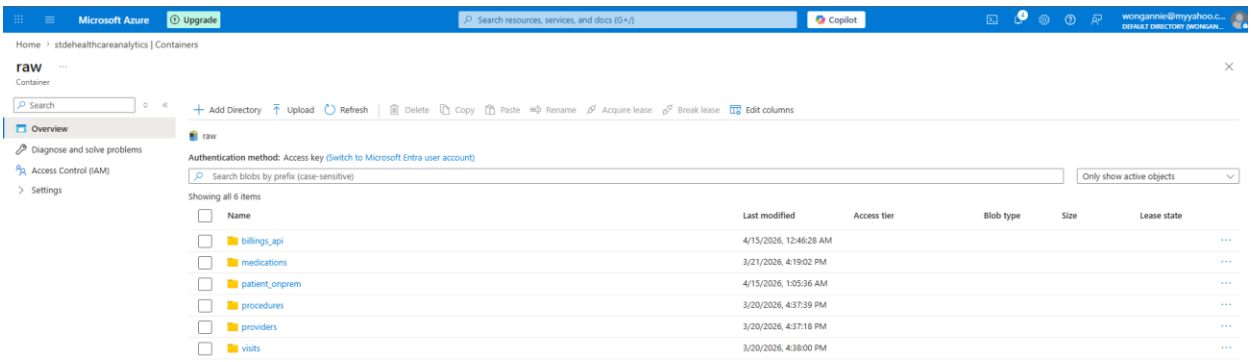
The architecture follows a modular and scalable design, where each dataset is processed independently but controlled through a master pipeline, ensuring maintainability, reusability, and efficient troubleshooting across the system.



Architecture diagram representing medallion pipeline with ADF orchestration, Databricks transformation, and SQL-based metadata-driven control, file tracking, and

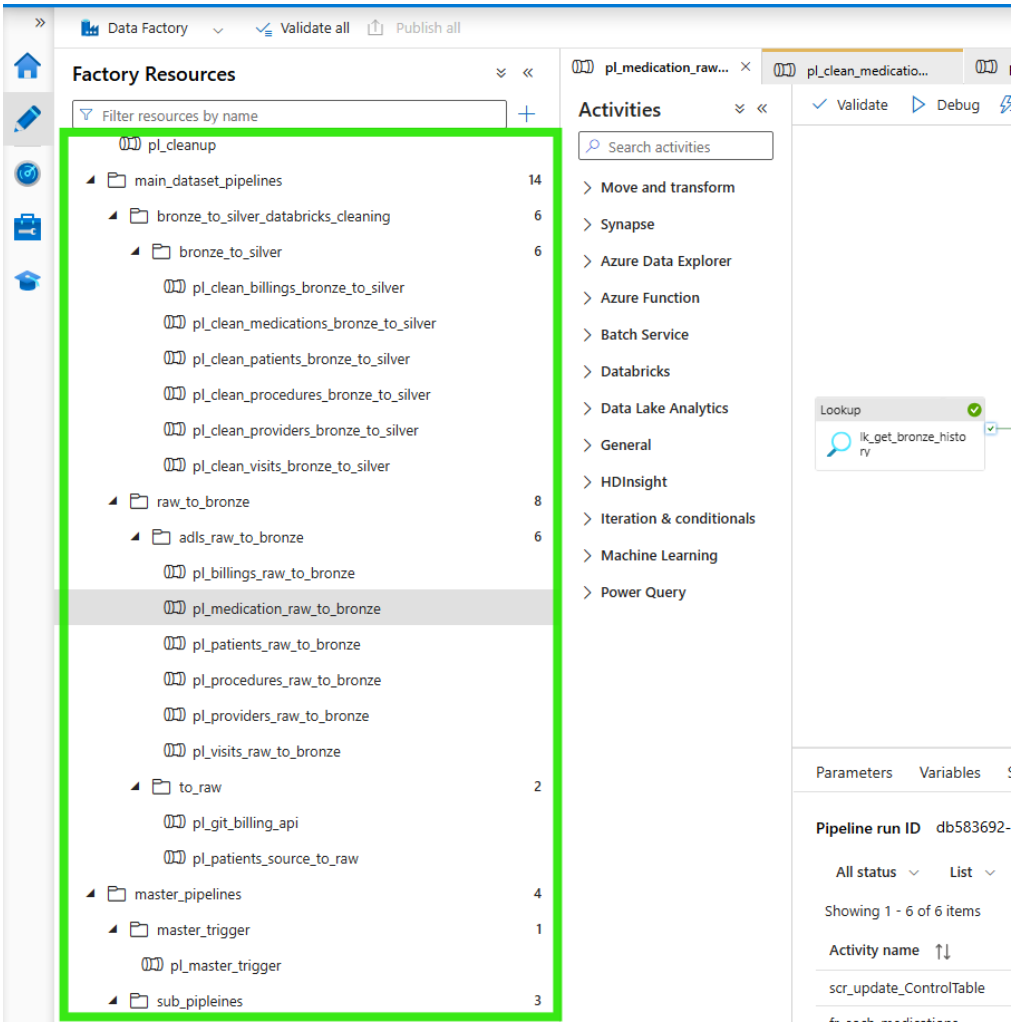


ADLS Gen2 storage structure showing container organization for Raw, Bronze, Silver, and Gold layers, enabling structured data storage and supporting the medallion



PIPELINE DESIGN

The pipeline was designed using a modular architecture, where each dataset is processed through separate pipelines for scalability and maintainability. A master pipeline orchestrates execution, managing dependencies and sequencing. Each pipeline includes file detection, ingestion, transformation, and audit logging. Conditional logic handles no-data scenarios to prevent failures. Databricks notebooks perform transformations in the Silver layer. Audit logs capture execution details for monitoring. The design ensures idempotent, rerunnable processing with clear visibility and robust error handling.



Azure Data Factory pipeline structure showing modular organization of pipelines for Source to Raw, Raw to Bronze, and Bronze to Silver layers, along with master

SQL SERVER CONTROL TABLE

The pipeline uses SQL tables to manage execution, track data processing, and maintain control over incremental loads. The Control Table stores metadata about each dataset, including source details, load type, and processing rules, enabling dynamic and scalable pipeline execution.

The Files Table tracks incoming files, capturing details such as file name, path, status, and processing timestamps. It helps ensure that only new or unprocessed files are picked up in each run, supporting incremental processing and preventing duplication.

The Audit Log table records pipeline execution details, including pipeline name, dataset, status, start time, end time, and error messages. It provides full visibility into pipeline runs, supports monitoring, and helps in troubleshooting and maintaining data reliability.

ControlTable

Data preview - ControlTable

Showing 1000 rows

	Id	TableName	LastProductionUpdateDate
1	1	Billings	2000-01-01T00:00:00.0000000
2	2	Patients	2026-04-15T20:20:53.3570000
3	3	Providers	2000-01-01T00:00:00.0000000
4	4	Visits	2000-01-01T00:00:00.0000000
5	5	Procedures	2000-01-01T00:00:00.0000000
6	6	Medications	2000-01-01T00:00:00.0000000

ControlTable: Tracks last processed timestamp for each dataset to enable incremental loading

AuditLog

Data preview - AuditLog

Showing 1000 rows

	Id	BatchName	TableName	Status	Layer	StartTime	EndTime
1	1	patients_batch2	Patients	Raw_Completed	Raw	2026-04-15T20:18:27.2070...	2026-04-15T20:19:04.5200...
2	2	patients_batch1	Patients	Raw_Completed	Raw	2026-04-15T20:18:49.2600...	2026-04-15T20:19:21.2330...
3	3	patients_batch3	Patients	Raw_Completed	Raw	2026-04-15T20:18:36.0230...	2026-04-15T20:19:05.3600...
4	4	patients_batch4	Patients	Raw_Completed	Raw	2026-04-15T20:18:26.6000...	2026-04-15T20:18:51.6530...
5	5	billing_batch1.json	Billing	Raw_Completed	Raw	2026-04-15T18:31:10.7900...	2026-04-15T20:17:04.5830...
6	6	providers_batch1	Providers	Success	Silver	2026-04-15T19:54:10.2700...	2026-04-15T19:55:03.9200...
7	7	procedures_batch1	Procedures	Success	Silver	2026-04-15T19:57:23.5930...	2026-04-15T19:58:02.4100...
8	8	medications_batch1	Medications	Success	Silver	2026-04-15T19:58:43.1330...	2026-04-15T19:59:51.3370...
9	9	patients_batch5	Patients	Success	Silver	2026-04-15T20:00:43.0130...	2026-04-15T20:01:50.8870...
10	10	billings_batch2	Billings	Success	Silver	2026-04-15T20:07:16.5230...	2026-04-15T20:08:20.3900...
11	11	visits_batch1	Visits	Success	Silver	2026-04-15T20:07:41.0900...	2026-04-15T20:08:52.7070...
12	12	billings_batch3	Billings	Success	Silver	2026-04-15T20:13:33.0470...	2026-04-15T20:14:22.0370...
13	13	providers_batch2	Providers	Success	Silver	2026-04-15T20:13:34.7300...	2026-04-15T20:14:32.4630...
14	14	procedures_batch2	Procedures	Success	Silver	2026-04-15T20:15:30.6800...	2026-04-15T20:16:42.3970...
15	15	medications_batch2	Medications	Success	Silver	2026-04-15T20:17:19.4130...	2026-04-15T20:18:28.3800...
16	16	providers_batch3	Providers	Success	Silver	2026-04-15T20:21:28.1100...	2026-04-15T20:22:17.9130...
17	17	procedures_batch3	Procedures	Success	Silver	2026-04-15T20:22:58.2900...	2026-04-15T20:23:42.8970...
18	18	medications_batch3	Medications	Success	Silver	2026-04-15T20:24:24.9970...	2026-04-15T20:25:23.7130...
19	19	visits_batch2	Visits	Success	Silver	2026-04-15T20:27:19.8070...	2026-04-15T20:28:22.6870...

AuditLog: Captures pipeline execution details including status, layer, and timing for monitoring and debugging

FilesTable

Data preview - FilesTable

Showing 1000 rows

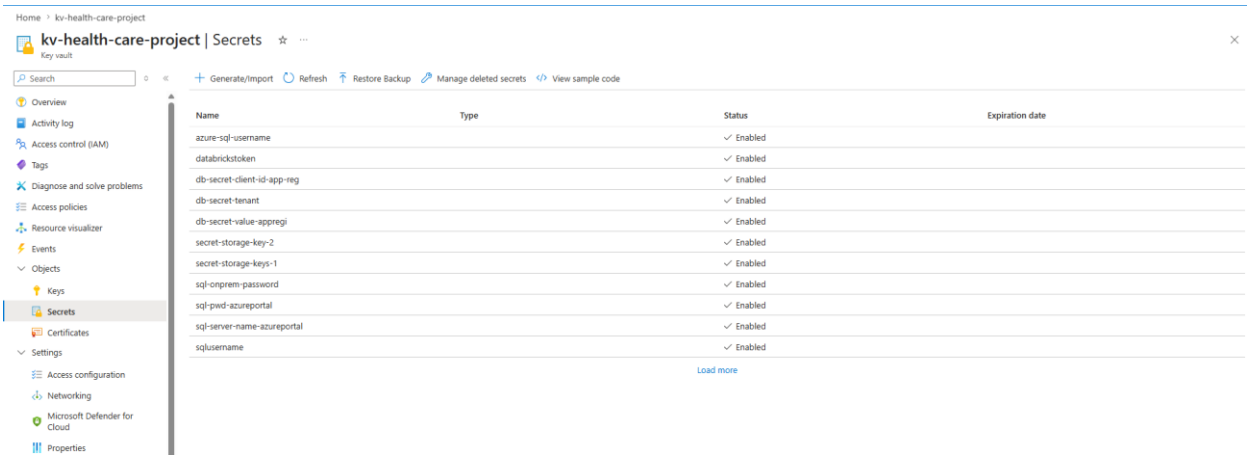
	Id	FileName	BatchName	Status	CreatedDate	UpdatedDate
1	1	providers_batch1	batch1	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T19:56:46.5670000
2	2	providers_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:14:52.2400000
3	3	providers_batch3	batch3	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:22:35.0400000
4	4	providers_batch4	batch4	Pending	2026-04-15T18:18:15.5500000	NULL
5	5	providers_batch5	batch5	Pending	2026-04-15T18:18:15.5500000	NULL
6	6	visits_batch1	batch1	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:09:02.3130000
7	7	visits_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:28:29.0330000
8	8	visits_batch3	batch3	Pending	2026-04-15T18:18:15.5500000	NULL
9	9	visits_batch4	batch4	Pending	2026-04-15T18:18:15.5500000	NULL
10	10	visits_batch5	batch5	Pending	2026-04-15T18:18:15.5500000	NULL
11	11	procedures_batch1	batch1	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T19:58:09.8470000
12	12	procedures_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:16:51.9300000
13	13	procedures_batch3	batch3	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:23:52.2500000
14	14	procedures_batch4	batch4	Pending	2026-04-15T18:18:15.5500000	NULL
15	15	procedures_batch5	batch5	Pending	2026-04-15T18:18:15.5500000	NULL
16	16	medications_batch1	batch1	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:00:00.7630000
17	17	medications_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:18:41.4130000
18	18	medications_batch3	batch3	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:25:33.5130000
19	19	medications_batch4	batch4	Pending	2026-04-15T18:18:15.5500000	NULL
20	20	medications_batch5	batch5	Pending	2026-04-15T18:18:15.5500000	NULL
21	21	billings_batch1	batch1	Raw_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:16:50.0470000
22	22	billings_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:08:39.2370000
23	23	billings_batch3	batch3	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:14:28.6730000
24	24	billings_batch4	batch4	Pending	2026-04-15T18:18:15.5500000	NULL
25	25	billings_batch5	batch5	Pending	2026-04-15T18:18:15.5500000	NULL
26	26	patients_batch1	batch1	Raw_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:19:32.9800000
27	27	patients_batch2	batch2	Raw_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:19:21.4100000
28	28	patients_batch3	batch3	Raw_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:19:14.3170000
29	29	patients_batch4	batch4	Raw_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:20:40.5200000
30	30	patients_batch5	batch5	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:02:09.4270000

FilesTable : manages file-level tracking and status to support dynamic, idempotent pipeline execution.

KEY VAULT

Azure Key Vault is used to securely store and manage sensitive information such as connection strings, database credentials, and access keys. It ensures that secrets are not hardcoded within the pipeline or notebooks.

The pipeline retrieves these secrets dynamically at runtime, enabling secure and centralized access to resources. This improves security, simplifies secret management, and supports best practices for production environments.

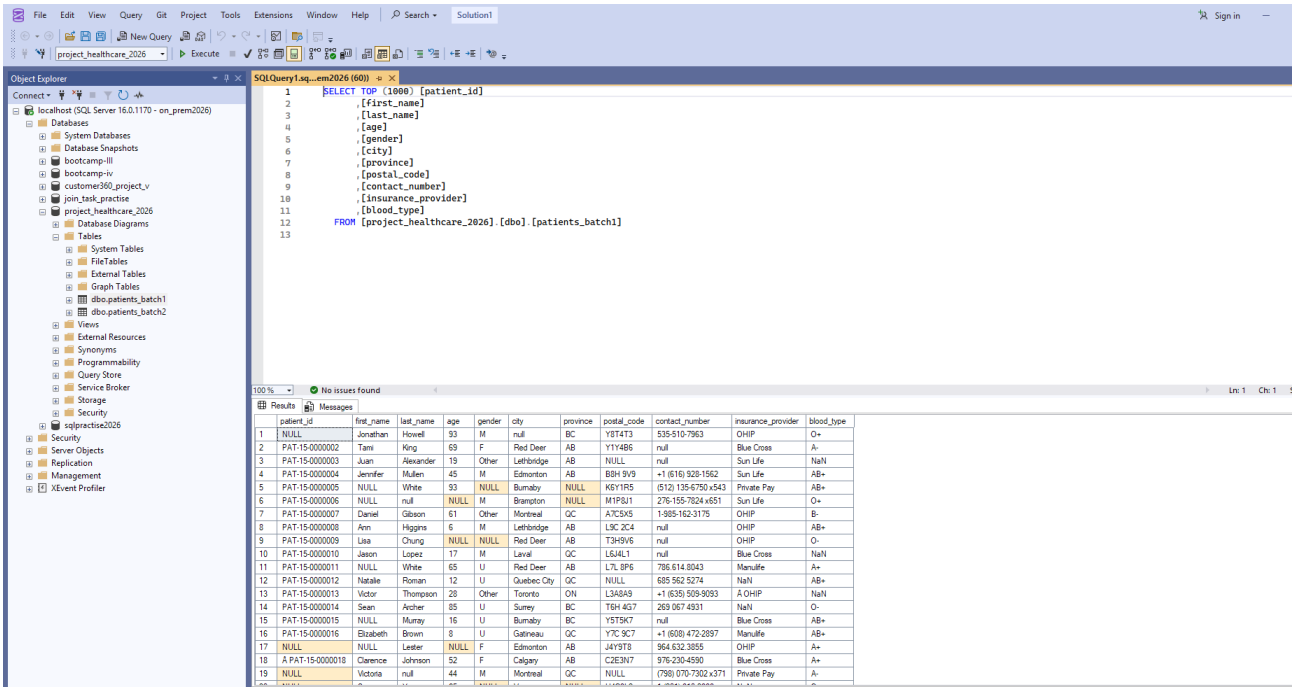


Azure Key Vault used for secure storage and management of secrets in the pipeline.

DATASET

We used synthetic healthcare datasets generated using Python to simulate a realistic data environment without exposing sensitive information. The datasets included Patients, Providers, Visits, Procedures, Medications, and Billings, each representing a key domain in a healthcare system. These datasets were interconnected using primary and foreign keys such as patient_id, provider_id, and visit_id, allowing us to model real-world relationships. The data initially contained inconsistencies like null values and duplicates, which helped us implement data cleaning and transformation logic. All datasets were ingested into the Raw layer, standardized in the Bronze layer, and transformed into analytics-ready formats in the Silver layer, supporting a complete end-to-end data engineering workflow.

SAMPLE SOURCE

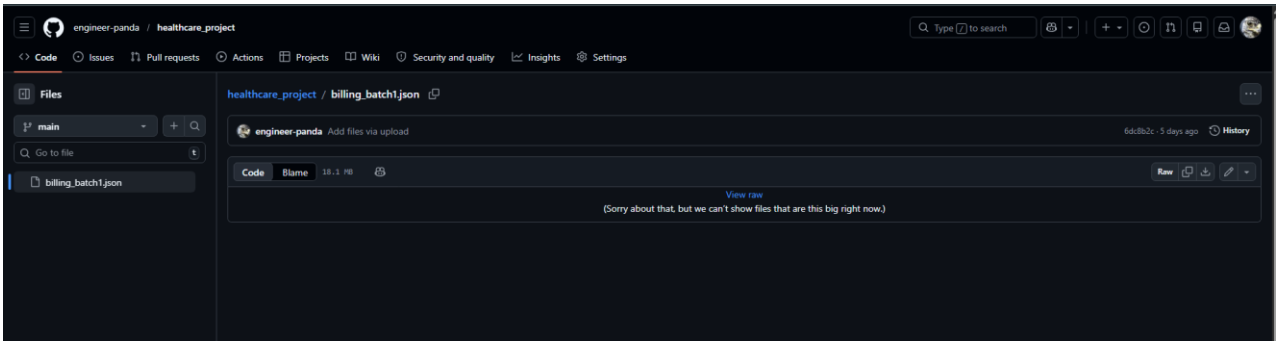


	patient_id	first_name	last_name	age	gender	city	province	postal_code	contact_number	insurance_provider	blood_type
1	NULL	Jonathan	Howell	93	M	null	BC	Y8T4T3	535-510-7963	OHIP	O+
2	PAT-15-0000002	Tami	King	69	F	Red Deer	AB	Y1Y4B6	null	Blue Cross	A-
3	PAT-15-0000003	Juan	Alexander	19	Other	Lethbridge	AB	NULL	null	Sun Life	NaN
4	PAT-15-0000004	Jennifer	Mullen	45	M	Edmonton	AB	B8H 9V9	+1 (616) 928-1562	Sun Life	AB+
5	PAT-15-0000005	NULL	White	93	NULL	Burnaby	NULL	K6Y1R5	(512) 135-6750 x543	Private Pay	AB+
6	PAT-15-0000006	NULL	null	NULL	M	Brampton	NULL	M1P8J1	276-155-7824 x651	Sun Life	O+
7	PAT-15-0000007	Daniel	Gibson	61	Other	Montreal	QC	A7C5X5	1-985-162-3175	OHIP	B-
8	PAT-15-0000008	Ann	Higgins	6	M	Lethbridge	AB	L9C 2C4	null	OHIP	AB+
9	PAT-15-0000009	Lisa	Chung	NULL	NULL	Red Deer	AB	T3H9V6	null	OHIP	O-
10	PAT-15-0000010	Jason	Lopez	17	M	Laval	QC	L6J4L1	null	Blue Cross	NaN
11	PAT-15-0000011	NULL	White	65	U	Red Deer	AB	L7L 8P6	786.614.8043	Manulife	A+
12	PAT-15-0000012	Natalie	Roman	12	U	Quebec City	QC	NULL	685 562 5274	NaN	AB+
13	PAT-15-0000013	Victor	Thompson	28	Other	Toronto	ON	L3A8A9	+1 (635) 509-9093	Â OHIP	NaN
14	PAT-15-0000014	Sean	Archer	85	U	Surrey	BC	T6H 4G7	269 067 4931	NaN	O-
15	PAT-15-0000015	NULL	Murray	16	U	Burnaby	BC	Y5T5K7	null	Blue Cross	AB+
16	PAT-15-0000016	Elizabeth	Brown	8	U	Gatineau	QC	Y7C 9C7	+1 (608) 472-2897	Manulife	AB+
17	NULL	NULL	Lester	NULL	F	Edmonton	AB	J4Y9T8	964.632.3855	OHIP	A+
18	Â PAT-15-0000018	Clarence	Johnson	52	F	Calgary	AB	C2E3N7	976-230-4590	Blue Cross	A+
19	NULL	Victoria	null	44	M	Montreal	QC	NULL	(798) 070-7302 x371	Private Pay	A-

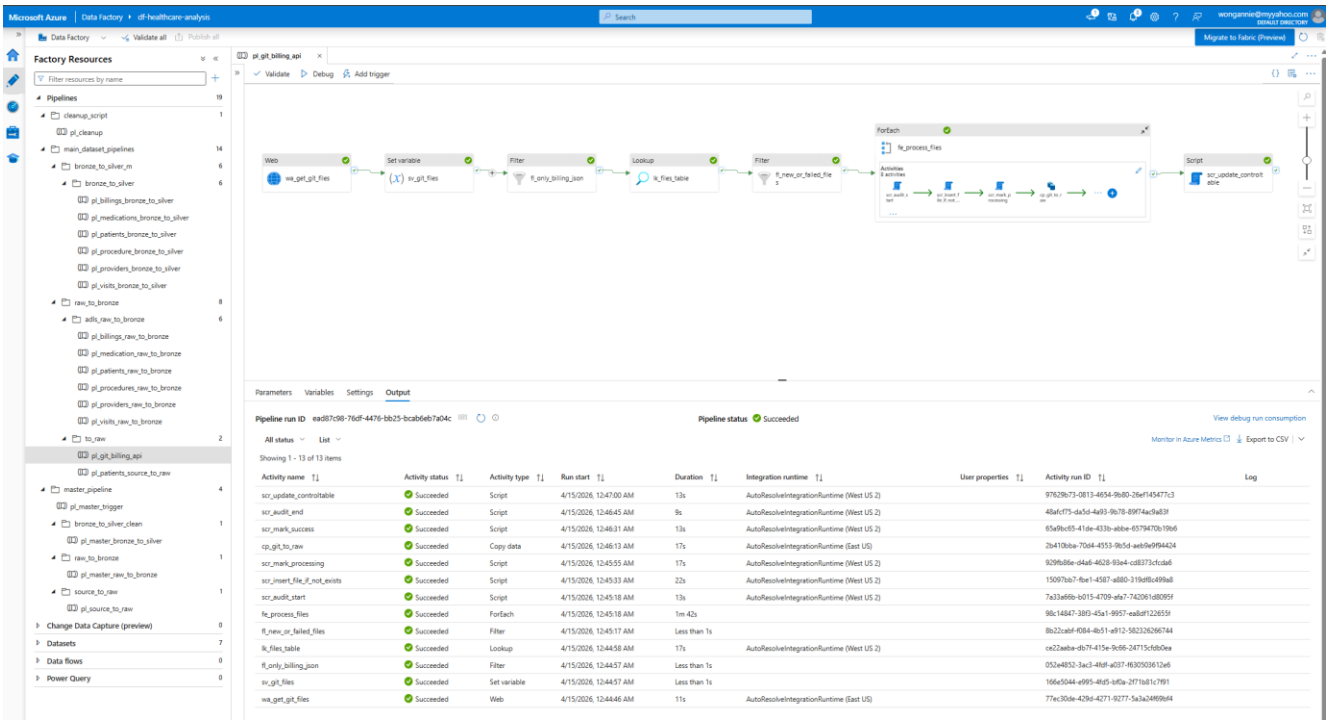
Sample synthetic Patients dataset in SQL Server used as source data for pipeline ingestion.

SOURCE TO RAW - BILLING – GITHUB HOST

We fetched the billing dataset from a GitHub-hosted source and ingested it into the ADLS Raw layer using Azure Data Factory. The data was stored in its original format without any transformations to preserve data integrity. This step ensured reliable ingestion, proper source connectivity, and a foundation for downstream processing in the Bronze and Silver layers.



GitHub used as source to host and ingest files into the data pipeline.



Ingested Patients data from on-prem SQL Server to ADLS Raw using Azure Integration Runtime and Copy Activity with incremental load.

AuditLog | ControlTable | FilesTable

Data preview - FilesTable

Showing 1000 rows

	Id	FileName	BatchName	Status	CreatedDate	UpdatedDate
1	1	providers_batch1	batch1	Pending	2026-04-15T04:24:55.5970000	NULL
2	2	providers_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
3	3	providers_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
4	4	providers_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
5	5	providers_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
6	6	visits_batch1	batch1	Pending	2026-04-15T04:24:55.5970000	NULL
7	7	visits_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
8	8	visits_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
9	9	visits_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
10	10	visits_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
11	11	procedures_batch1	batch1	Pending	2026-04-15T04:24:55.5970000	NULL
12	12	procedures_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
13	13	procedures_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
14	14	procedures_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
15	15	procedures_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
16	16	medications_batch1	batch1	Pending	2026-04-15T04:24:55.5970000	NULL
17	17	medications_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
18	18	medications_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
19	19	medications_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
20	20	medications_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
21	21	billings_batch1	batch1	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:06:44.8200000
22	22	billings_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
23	23	billings_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
24	24	billings_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
25	25	billings_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
26	26	patients_batch1	batch1	Raw_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:06:05.2470000
27	27	patients_batch2	batch2	Raw_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:06:43.2330000
28	28	patients_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
29	29	patients_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
30	30	patients_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL

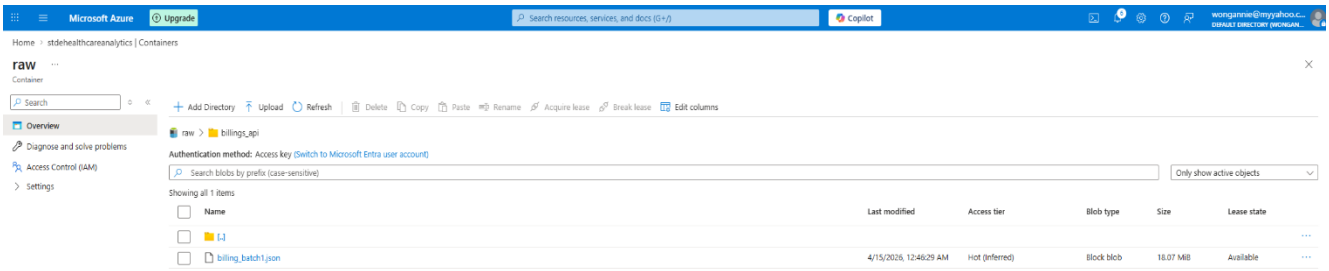
AuditLog

Data preview - AuditLog

Showing 1000 rows

	Id	BatchName	TableName	Status	Layer	StartTime	EndTime
1	1	billing_batch1.json	Billing	Raw_Completed	Raw	2026-04-15T04:45:30.2970000	2026-04-15T04:46:51.9300000
2	2	patients_batch1	Patients	Raw_Completed	Raw	2026-04-15T05:05:23.8500000	2026-04-15T05:05:54.6170000
3	3	patients_batch2	Patients	Raw_Completed	Raw	2026-04-15T05:05:19.4370000	2026-04-15T05:05:51.0400000
4	4	billing_batch1	Billings	Bronze_Completed	Bronze	2026-04-15T05:05:59.6800000	2026-04-15T05:07:07.8470000

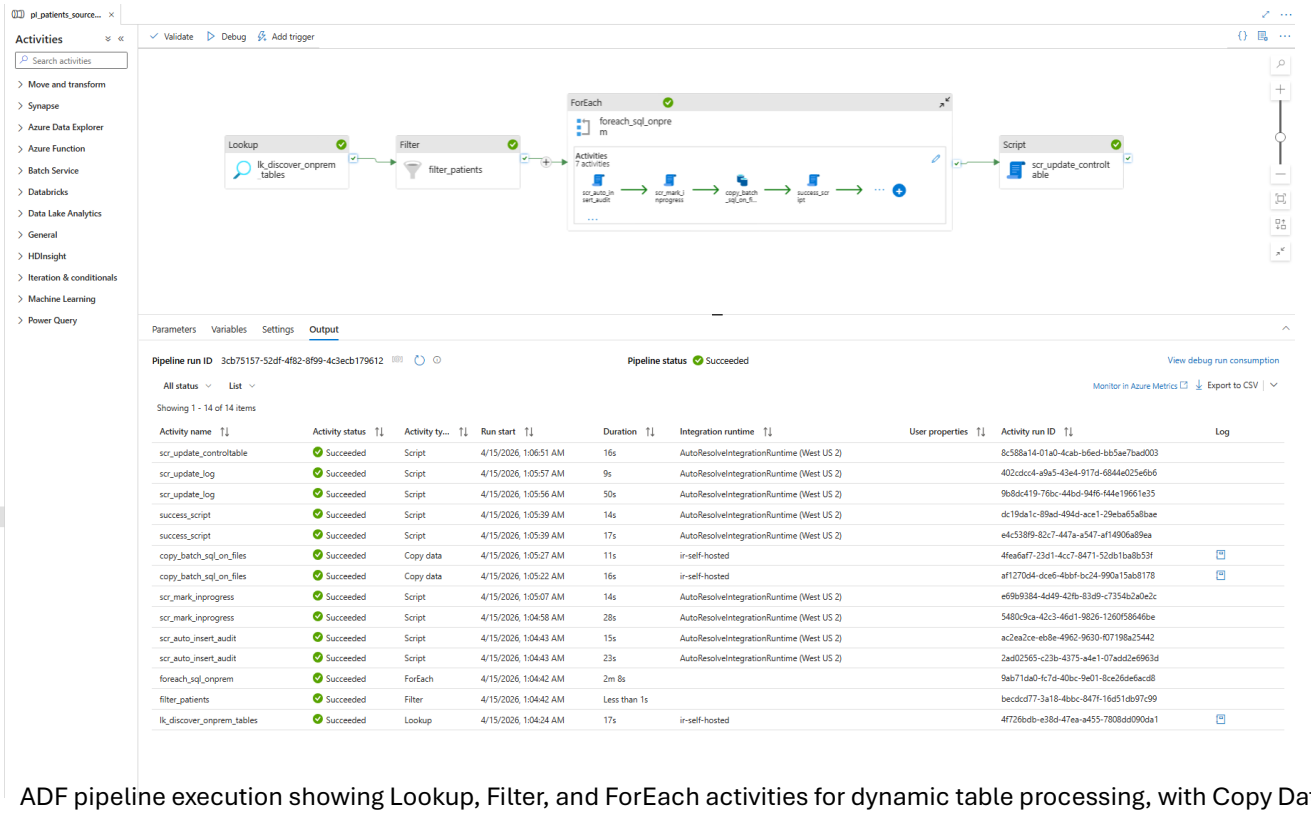
Data ingested using Integration Runtime and Copy activity, with FilesTable and AuditLog updated for tracking and monitoring.



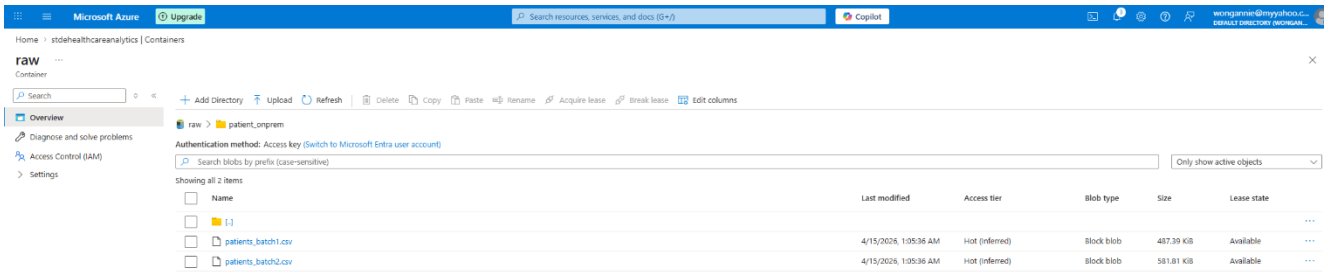
Source data ingested into “Raw layer” using ADF Copy activity.

SOURCE TO RAW – PATIENTS – ON- PREM SQL SSMS SERVER

We connected to the on-premises SQL Server (SSMS) using a Self-Hosted Integration Runtime and configured a Linked Service in Azure Data Factory. Using a Copy Data activity, we performed incremental ingestion of the Patients dataset based on a watermark column. The data was read and stored in the ADLS Raw layer without transformation. Additionally, audit logging captured pipeline start time, end time, and execution status for monitoring and traceability.



ADF pipeline execution showing Lookup, Filter, and ForEach activities for dynamic table processing, with Copy Data and Script activities successfully ingesting data and updating ControlTable and AuditLog for tracking and monitoring.



Patients dataset successfully ingested into ADLS Raw layer as CSV files using ADF Copy activity

AuditLog | ControlTable | FilesTable

Data preview - FilesTable

Showing 1000 rows

	Id	FileName	BatchName	Status	CreatedDate	UpdatedDate
1	1	providers_batch1	batch1	Pending	2026-04-15T04:24:55.5970000	NULL
2	2	providers_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
3	3	providers_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
4	4	providers_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
5	5	providers_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
6	6	visits_batch1	batch1	Pending	2026-04-15T04:24:55.5970000	NULL
7	7	visits_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
8	8	visits_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
9	9	visits_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
10	10	visits_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
11	11	procedures_batch1	batch1	Pending	2026-04-15T04:24:55.5970000	NULL
12	12	procedures_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
13	13	procedures_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
14	14	procedures_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
15	15	procedures_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
16	16	medications_batch1	batch1	Pending	2026-04-15T04:24:55.5970000	NULL
17	17	medications_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
18	18	medications_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
19	19	medications_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
20	20	medications_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
21	21	billings_batch1	batch1	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:06:44.8200000
22	22	billings_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
23	23	billings_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
24	24	billings_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
25	25	billings_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
26	26	patients_batch1	batch1	Raw_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:06:05.2470000
27	27	patients_batch2	batch2	Raw_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:06:43.2330000
28	28	patients_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
29	29	patients_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
30	30	patients_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL

FilesTable updated after ingestion, showing file-level status changes (e.g., Pending to Raw_Processed), enabling tracking of processed batches and supporting controlled, idempotent pipeline execution.

AuditLog | ControlTable | FilesTable

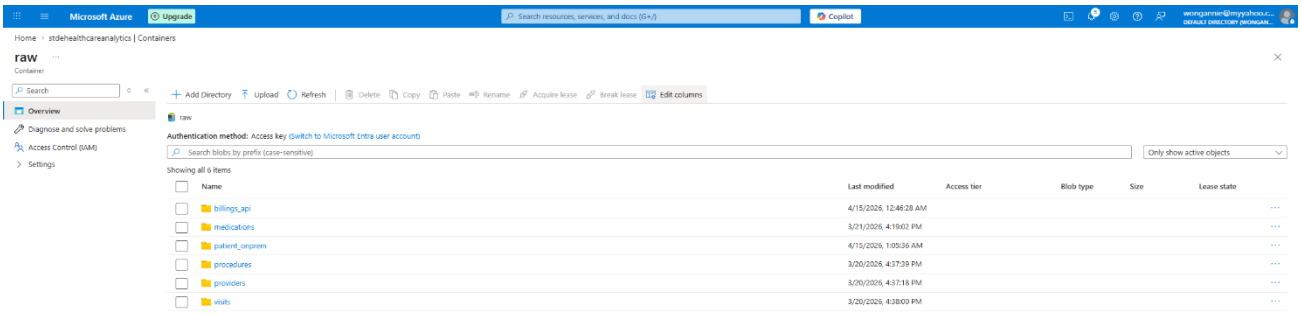
Data preview - AuditLog

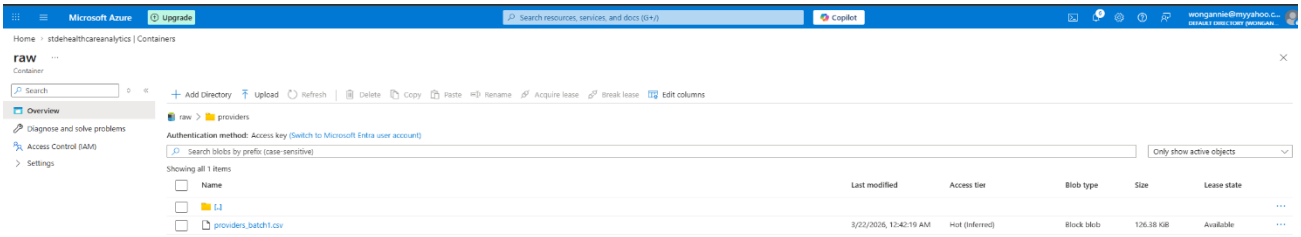
Showing 1000 rows

	Id	BatchName	TableName	Status	Layer	StartTime	EndTime
1	1	billing_batch1.json	Billing	Raw_Completed	Raw	2026-04-15T04:45:30.2970000	2026-04-15T04:46:51.9300000
2	2	patients_batch1	Patients	Raw_Completed	Raw	2026-04-15T05:05:23.8500000	2026-04-15T05:05:54.6170000
3	3	patients_batch2	Patients	Raw_Completed	Raw	2026-04-15T05:05:19.4370000	2026-04-15T05:05:51.0400000
4	4	billing_batch1	Billings	Bronze_Completed	Bronze	2026-04-15T05:05:59.6800000	2026-04-15T05:07:07.8470000

AuditLog capturing pipeline execution details including batch name, table name, layer, start time, end time, and status, providing visibility and monitoring for each pipeline run.

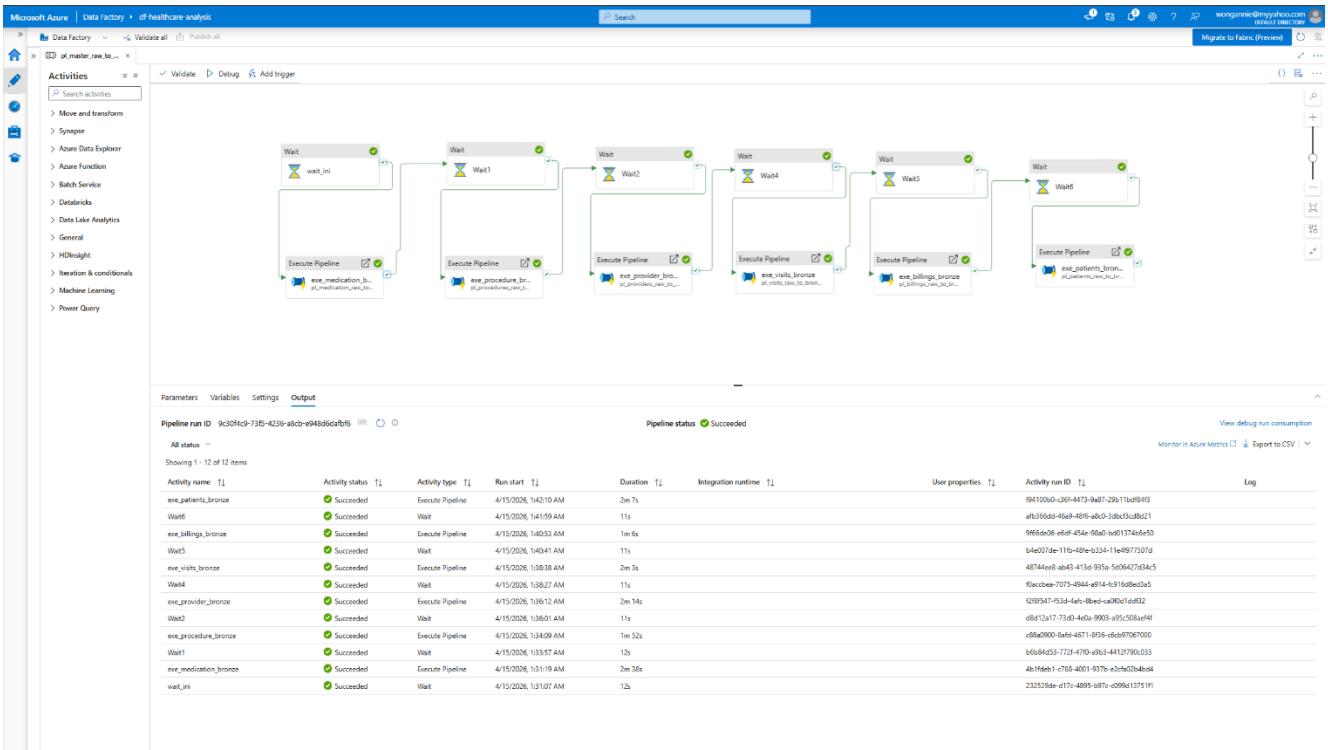
RAW - PROVIDERS – ADLS



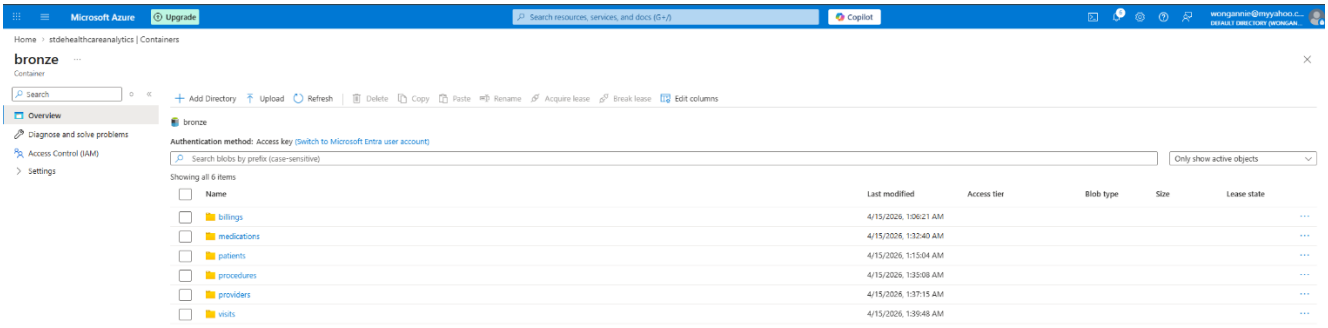


Used ADLS Storage Account Raw container to store Providers dataset after ingestion from source.

MASTER RAW TO BRONZE :



Master Pipeline run completed successfully, invoking all Raw layer pipelines. All activities executed without errors, ensuring data ingestion, processing, and metadata updates were performed as expected.



Used ADLS Storage Account Raw container to store Providers dataset after ingestion from source.

FilesTable

Data preview - FilesTable

Showing 1000 rows

	Id	FileName	BatchName	Status	CreatedDate	UpdatedDate
1	1	providers_batch1	batch1	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
2	2	providers_batch2	batch2	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
3	3	providers_batch3	batch3	Pending	2026-04-15T22:34:52.4830000	NULL
4	4	providers_batch4	batch4	Pending	2026-04-15T22:34:52.4830000	NULL
5	5	providers_batch5	batch5	Pending	2026-04-15T22:34:52.4830000	NULL
6	6	visits_batch1	batch1	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
7	7	visits_batch2	batch2	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
8	8	visits_batch3	batch3	Pending	2026-04-15T22:34:52.4830000	NULL
9	9	visits_batch4	batch4	Pending	2026-04-15T22:34:52.4830000	NULL
10	10	visits_batch5	batch5	Pending	2026-04-15T22:34:52.4830000	NULL
11	11	procedures_batch1	batch1	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
12	12	procedures_batch2	batch2	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
13	13	procedures_batch3	batch3	Pending	2026-04-15T22:34:52.4830000	NULL
14	14	procedures_batch4	batch4	Pending	2026-04-15T22:34:52.4830000	NULL
15	15	procedures_batch5	batch5	Pending	2026-04-15T22:34:52.4830000	NULL
16	16	medications_batch1	batch1	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
17	17	medications_batch2	batch2	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
18	18	medications_batch3	batch3	Pending	2026-04-15T22:34:52.4830000	NULL
19	19	medications_batch4	batch4	Pending	2026-04-15T22:34:52.4830000	NULL
20	20	medications_batch5	batch5	Pending	2026-04-15T22:34:52.4830000	NULL
21	21	billings_batch1	batch1	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
22	22	billings_batch2	batch2	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
23	23	billings_batch3	batch3	Pending	2026-04-15T22:34:52.4830000	NULL
24	24	billings_batch4	batch4	Pending	2026-04-15T22:34:52.4830000	NULL
25	25	billings_batch5	batch5	Pending	2026-04-15T22:34:52.4830000	NULL
26	26	patients_batch1	batch1	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
27	27	patients_batch2	batch2	Raw_Completed	2026-04-15T22:34:52.4830000	2026-04-15T22:38:54.4470000
28	28	patients_batch3	batch3	Raw_Processed	2026-04-15T22:34:52.4830000	2026-04-15T22:35:16.5800000
29	29	patients_batch4	batch4	Pending	2026-04-15T22:34:52.4830000	NULL
30	30	patients_batch5	batch5	Pending	2026-04-15T22:34:52.4830000	NULL

FilesTable showing batch-level tracking of datasets with status updates (Pending to Raw_Completed), enabling controlled ingestion, monitoring of processed files, and support for idempotent pipeline execution.

Data preview - AuditLog

Showing 1000 rows

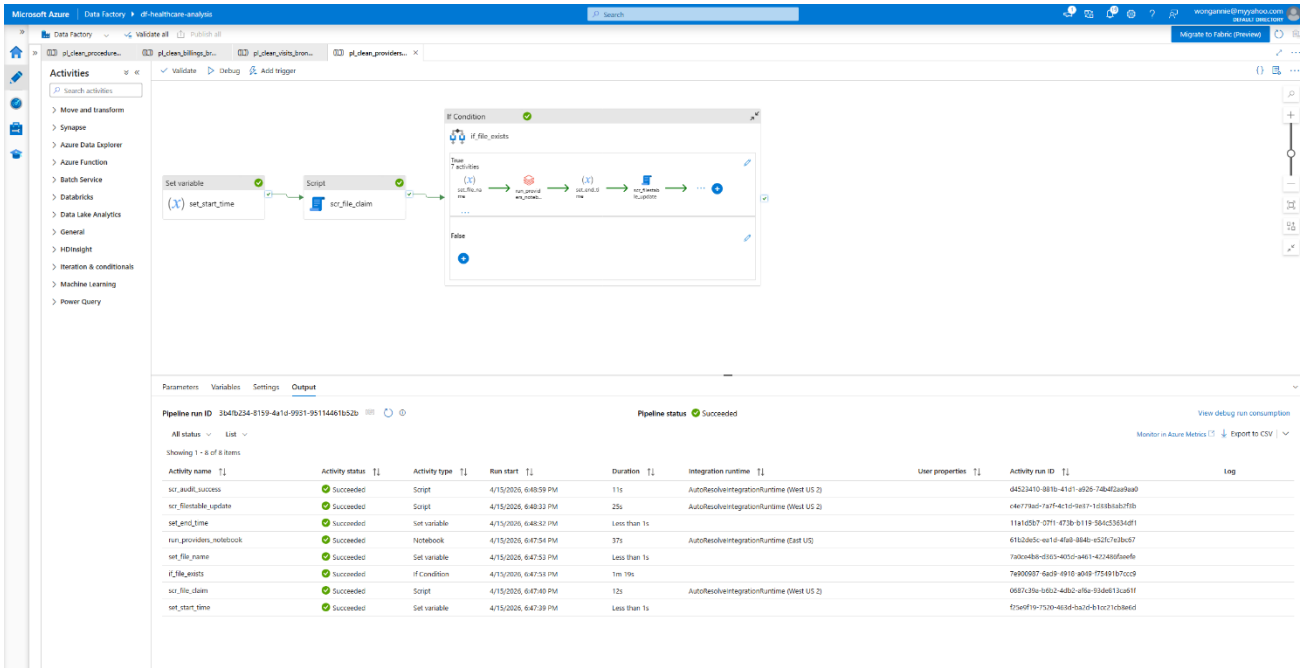
	Id	BatchName	TableName	Status	Layer	StartTime	EndTime
1	1	visits_batch1	Visits	Bronze_Processed	Bronze	2026-04-15T23:16:22.7730000	2026-04-15T23:23:42.0100000
2	2	procedures_batch1	Procedures	Bronze_Completed	Bronze	2026-04-15T23:28:20.3200000	2026-04-15T23:28:55.3470000
3	3	providers_batch1	Providers	Bronze_Processed	Bronze	2026-04-15T23:18:09.9170000	2026-04-15T23:23:42.0100000
4	4	visits_batch2	Visits	Bronze_Processed	Bronze	2026-04-15T23:18:42.6530000	2026-04-15T23:23:42.0100000
5	5	providers_batch1	Providers	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
6	6	providers_batch2	Providers	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
7	7	providers_batch3	Providers	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
8	8	visits_batch1	Visits	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
9	9	visits_batch2	Visits	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
10	10	visits_batch3	Visits	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
11	11	procedures_batch1	Procedures	Bronze_Completed	Bronze	2026-04-15T23:28:20.3200000	2026-04-15T23:28:55.3470000
12	12	procedures_batch2	Procedures	Bronze_Completed	Bronze	2026-04-15T23:28:18.8830000	2026-04-15T23:29:00.1200000
13	13	procedures_batch3	Procedures	Bronze_Completed	Bronze	2026-04-15T23:28:18.8170000	2026-04-15T23:28:49.2670000
14	14	medications_batch1	Medications	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
15	15	medications_batch2	Medications	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
16	16	medications_batch3	Medications	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
17	17	billings_batch1	Billings	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
18	18	billings_batch2	Billings	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
19	19	billings_batch3	Billings	Bronze_Processed	Bronze	2026-04-15T23:19:32.3670000	2026-04-15T23:23:42.0100000
20	23	providers_batch2	Providers	Success	Silver	2026-04-15T23:23:11.7200000	2026-04-15T23:24:00.9170000
21	24	medications_batch2.csv	Medications	Bronze_Completed	Bronze	2026-04-15T23:25:55.1230000	2026-04-15T23:26:28.2900000
22	25	medications_batch1.csv	Medications	Bronze_Completed	Bronze	2026-04-15T23:26:03.2930000	2026-04-15T23:26:49.0730000
23	26	patients_batch4	Patients	Raw_Completed	Raw	2026-04-15T23:30:49.2170000	2026-04-15T23:31:19.2700000
24	27	patients_batch1	Patients	Raw_Completed	Raw	2026-04-15T23:30:50.8800000	2026-04-15T23:31:24.8100000
25	28	patients_batch2	Patients	Raw_Completed	Raw	2026-04-15T23:30:48.5400000	2026-04-15T23:31:21.3830000
26	29	patients_batch3	Patients	Raw_Completed	Raw	2026-04-15T23:30:54.7800000	2026-04-15T23:31:26.7530000
27	30	billing_batch1.json	Billing	InProgress	Raw	2026-04-15T23:30:54.6230000	2026-04-15T23:32:03.3970000

FilesTable showing batch-level tracking of datasets with status updates (Pending to Raw_Processed), enabling controlled ingestion, monitoring of processed files, and support for idempotent pipeline execution.

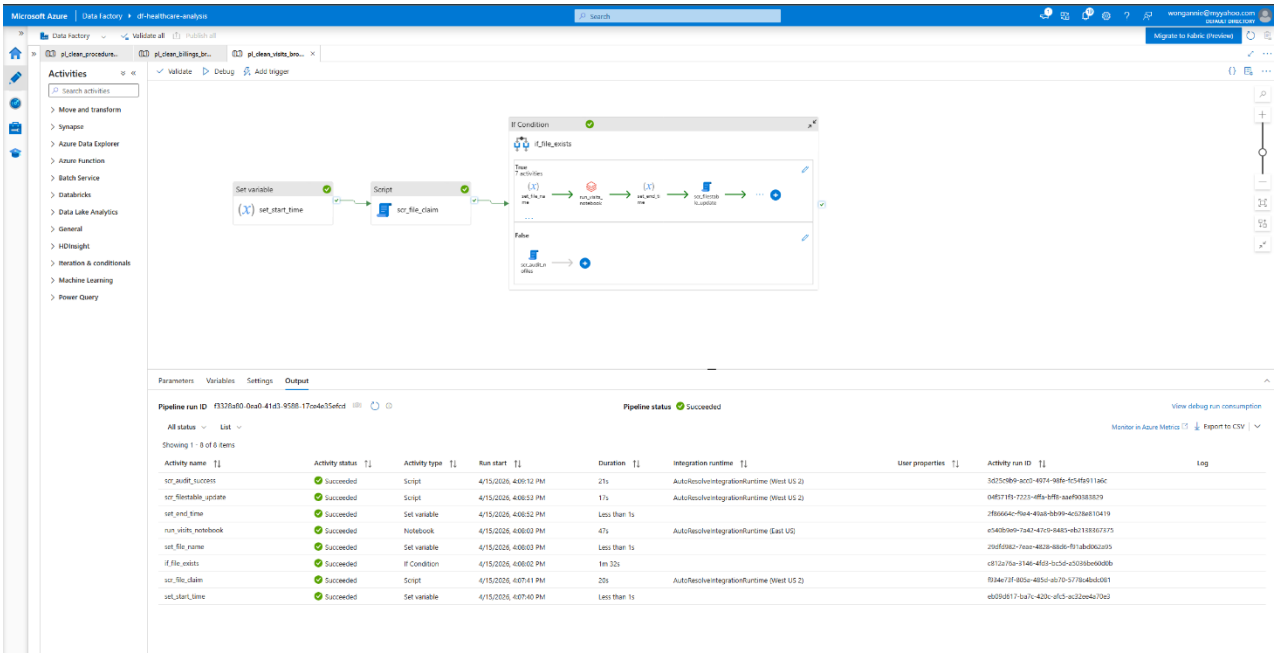
BRONZE TO SILVER

This pipeline processes the Providers dataset from the Bronze layer to the Silver layer by executing Databricks notebooks. It performs data cleaning, validation, and deduplication to ensure high-quality and consistent data. Business rules are applied to standardize fields and remove invalid records. The pipeline ensures that no file is processed more than once by tracking file status in the FilesTable. It also manages execution flow and dependencies to maintain proper sequencing of activities. Additionally, the AuditLog table is updated with details such as start time, end time, and execution status, providing full visibility and monitoring. This design ensures reliable, scalable, and idempotent data processing suitable for production environments.

The below screenshots show two Silver layer cleaning pipelines with Databricks integration, where data is transformed from Bronze to Silver using notebooks, applying cleaning, validation, and deduplication while updating FilesTable and AuditLog for tracking and monitoring.



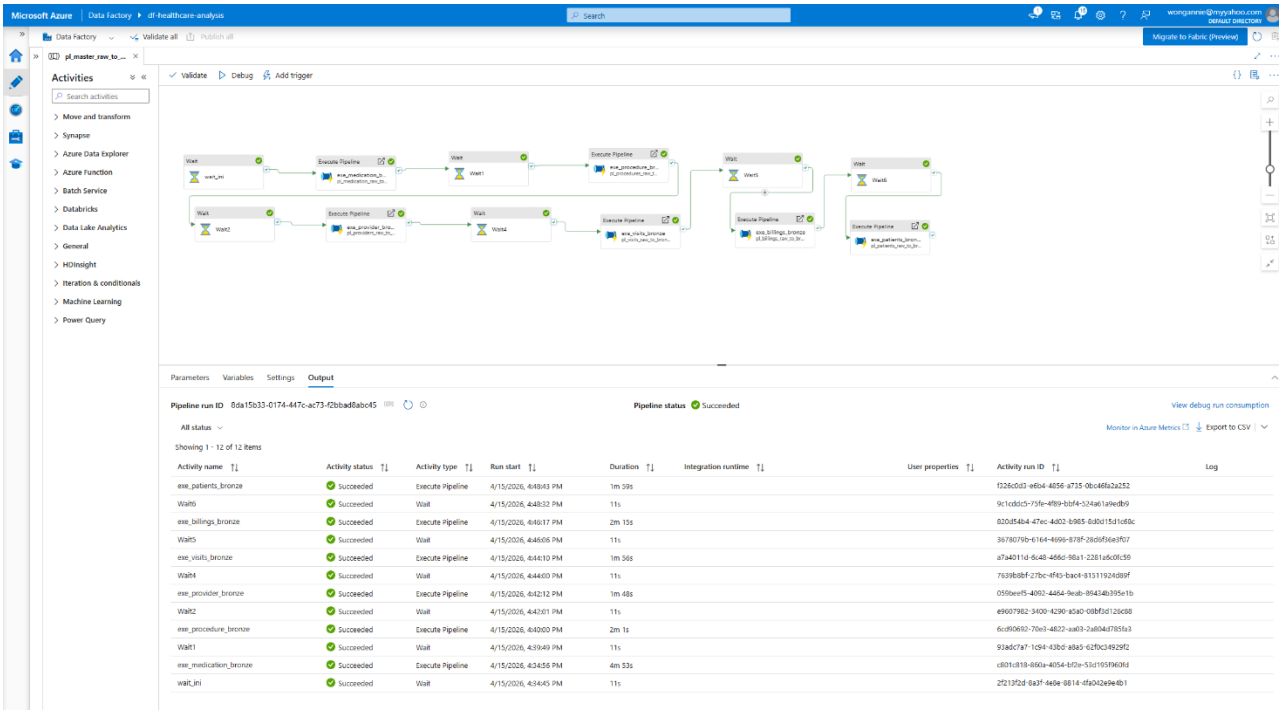
Pipeline execution showing conditional logic to check file availability before processing. If a file exists, it triggers Databricks notebook execution for transformation, updates FilesTable status, and logs execution details in AuditLog, ensuring controlled and reliable processing.



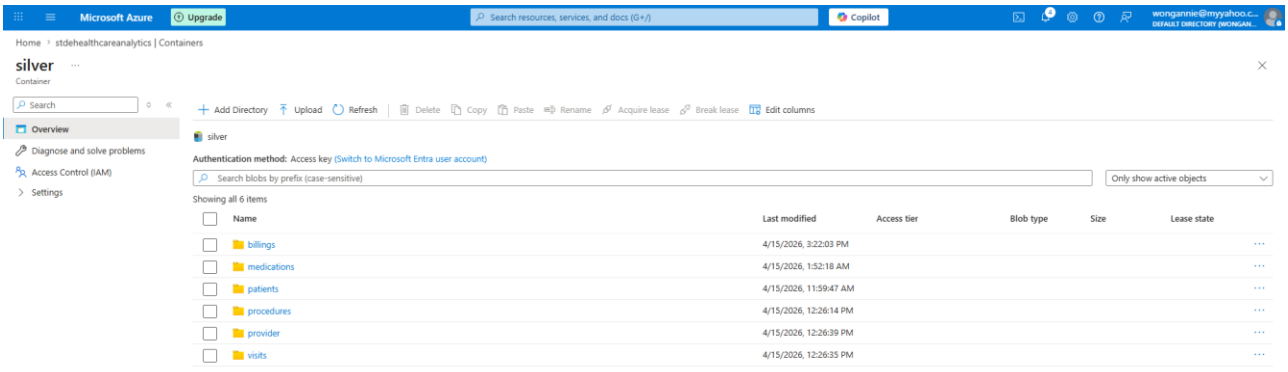
Visits Bronze to Silver pipeline execution showing conditional file check, followed by Databricks notebook run for data transformation. The pipeline performs cleaning, validation, and deduplication, updates FilesTable to prevent reprocessing, and logs execution details in AuditLog for monitoring and reliability.

MASTER BRONZE TO SILVER

The below screenshot shows the Master Pipeline orchestrating execution of all dataset pipelines across layers. It triggers individual pipelines using Execute Pipeline activities, manages dependencies and sequencing, and ensures end-to-end data processing while maintaining tracking through FilesTable and AuditLog.



Bronze to Silver Master Pipeline run completed successfully, orchestrating all dataset pipelines for transformation using Databricks, with proper sequencing and dependency handling, ensuring data is cleaned, validated, and loaded into the Silver layer.



FilesTable

Data preview - FilesTable

Showing 1000 rows

	Id	FileName	BatchName	Status	CreatedDate	UpdatedDate
1	1	providers_batch1	batch1	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
2	2	providers_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:14:52.2400000
3	3	providers_batch3	batch3	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:22:35.0400000
4	4	providers_batch4	batch4	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:50:32.8630000
5	5	providers_batch5	batch5	Pending	2026-04-15T18:18:15.5500000	NULL
6	6	visits_batch1	batch1	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
7	7	visits_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
8	8	visits_batch3	batch3	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:57:11.6100000
9	9	visits_batch4	batch4	Pending	2026-04-15T18:18:15.5500000	NULL
10	10	visits_batch5	batch5	Pending	2026-04-15T18:18:15.5500000	NULL
11	11	procedures_batch1	batch1	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
12	12	procedures_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
13	13	procedures_batch3	batch3	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
14	14	procedures_batch4	batch4	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:52:11.7700000
15	15	procedures_batch5	batch5	Pending	2026-04-15T18:18:15.5500000	NULL
16	16	medications_batch1	batch1	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
17	17	medications_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
18	18	medications_batch3	batch3	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:25:33.5130000
19	19	medications_batch4	batch4	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:53:37.6570000
20	20	medications_batch5	batch5	Pending	2026-04-15T18:18:15.5500000	NULL
21	21	billings_batch1	batch1	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
22	22	billings_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:08:39.2370000
23	23	billings_batch3	batch3	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
24	24	billings_batch4	batch4	Pending	2026-04-15T18:18:15.5500000	NULL
25	25	billings_batch5	batch5	Pending	2026-04-15T18:18:15.5500000	NULL
26	26	patients_batch1	batch1	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
27	27	patients_batch2	batch2	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
28	28	patients_batch3	batch3	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
29	29	patients_batch4	batch4	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T21:48:58.6100000
30	30	patients_batch5	batch5	Silver_Processed	2026-04-15T18:18:15.5500000	2026-04-15T20:02:09.4270000

FilesTable showing final status updates with datasets marked as Silver_Processed, indicating successful completion of Bronze to Silver transformations, with tracking of batch processing and ensuring no duplicate file execution.

AuditLog

Data preview - AuditLog

Showing 1000 rows

	Id	BatchName	TableName	Status	Layer	StartTime	EndTime
1	1	providers_batch1	Providers	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
2	2	providers_batch2	Providers	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
3	3	providers_batch3	Providers	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
4	4	visits_batch1	Visits	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
5	5	visits_batch2	Visits	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
6	6	visits_batch3	Visits	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
7	7	procedures_batch1	Procedures	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
8	8	procedures_batch2	Procedures	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
9	9	procedures_batch3	Procedures	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
10	10	medications_batch1	Medications	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
11	11	medications_batch2	Medications	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
12	12	medications_batch3	Medications	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
13	13	billings_batch1	Billings	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
14	14	billings_batch2	Billings	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000
15	15	billings_batch3	Billings	Silver_Processed	Silver	2026-04-15T23:12:34.7000000	2026-04-15T23:12:34.7000000

AuditLog showing successful execution of pipelines with status marked as Success for Silver layer processing, capturing batch name, dataset, start time, and end time, providing full visibility into end-to-end pipeline execution.

ERROR HANDLING AND LOGGING

The pipeline is designed to handle errors gracefully and ensure proper logging at each stage of execution. Validation checks are implemented to prevent failures due to missing or empty data, and conditional logic is used to handle no-data scenarios without breaking the pipeline.

Each activity is configured with success and failure paths to capture execution status. In case of failure, appropriate audit logs are recorded with error details to support troubleshooting.

Audit tables maintain records of pipeline runs, including start time, end time, status, and processed datasets. This ensures full visibility, traceability, and easier debugging in production environments.

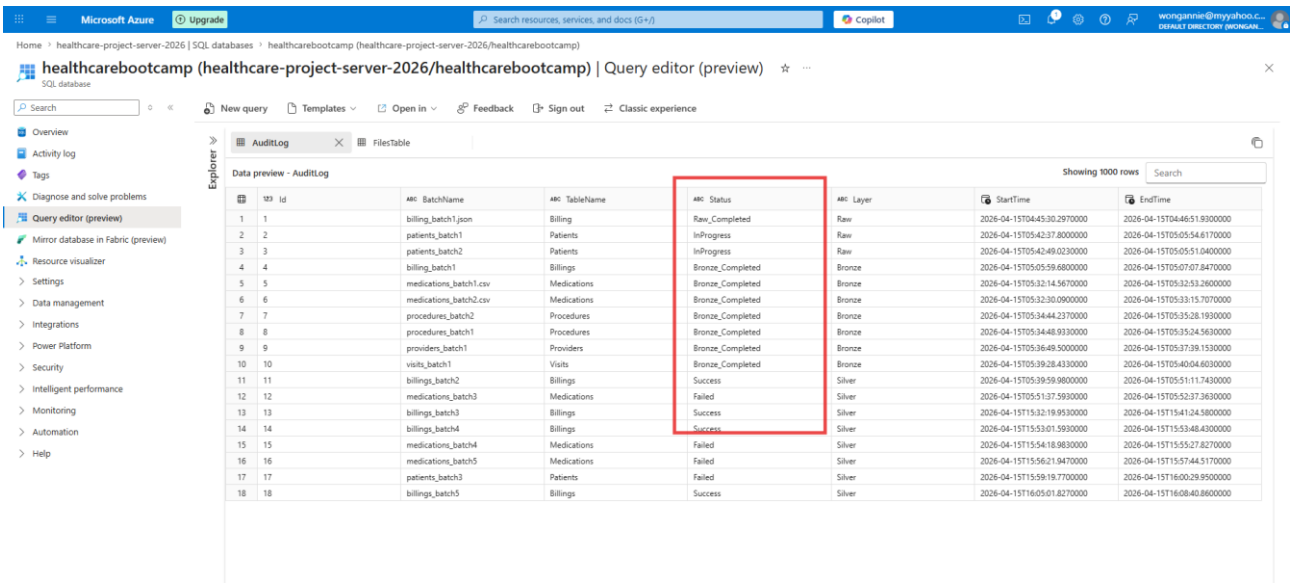
AuditLog | FilesTable

Data preview - FilesTable

Showing 1000 rows

	Id	FileName	BatchName	Status	CreatedDate	UpdatedDate
1	1	providers_batch1	batch1	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:38:08.8400000
2	2	providers_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
3	3	providers_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
4	4	providers_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
5	5	providers_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
6	6	visits_batch1	batch1	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:40:15.4500000
7	7	visits_batch2	batch2	Pending	2026-04-15T04:24:55.5970000	NULL
8	8	visits_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
9	9	visits_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
10	10	visits_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
11	11	procedures_batch1	batch1	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:35:41.8100000
12	12	procedures_batch2	batch2	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:35:39.2800000
13	13	procedures_batch3	batch3	Pending	2026-04-15T04:24:55.5970000	NULL
14	14	procedures_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
15	15	procedures_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL
16	16	medications_batch1	batch1	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:33:28.0930000
17	17	medications_batch2	batch2	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:33:27.5600000
18	18	medications_batch3	batch3	Failed	2026-04-15T04:24:55.5970000	2026-04-15T05:52:58.5270000
19	19	medications_batch4	batch4	Failed	2026-04-15T04:24:55.5970000	2026-04-15T15:55:45.6330000
20	20	medications_batch5	batch5	Failed	2026-04-15T04:24:55.5970000	2026-04-15T15:57:50.7470000
21	21	billings_batch1	batch1	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:06:44.8200000
22	22	billings_batch2	batch2	Silver_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:51:19.8400000
23	23	billings_batch3	batch3	Silver_Processed	2026-04-15T04:24:55.5970000	2026-04-15T15:41:41.7430000
24	24	billings_batch4	batch4	Silver_Processed	2026-04-15T04:24:55.5970000	2026-04-15T15:53:54.4270000
25	25	billings_batch5	batch5	Silver_Processed	2026-04-15T04:24:55.5970000	2026-04-15T16:08:53.0070000
26	26	patients_batch1	batch1	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:43:28.0070000
27	27	patients_batch2	batch2	Bronze_Processed	2026-04-15T04:24:55.5970000	2026-04-15T05:43:40.1270000
28	28	patients_batch3	batch3	Failed	2026-04-15T04:24:55.5970000	2026-04-15T16:00:40.0830000
29	29	patients_batch4	batch4	Pending	2026-04-15T04:24:55.5970000	NULL
30	30	patients_batch5	batch5	Pending	2026-04-15T04:24:55.5970000	NULL

FilesTable showing failed medication batches (batch3–batch5) while earlier batches were successfully processed.

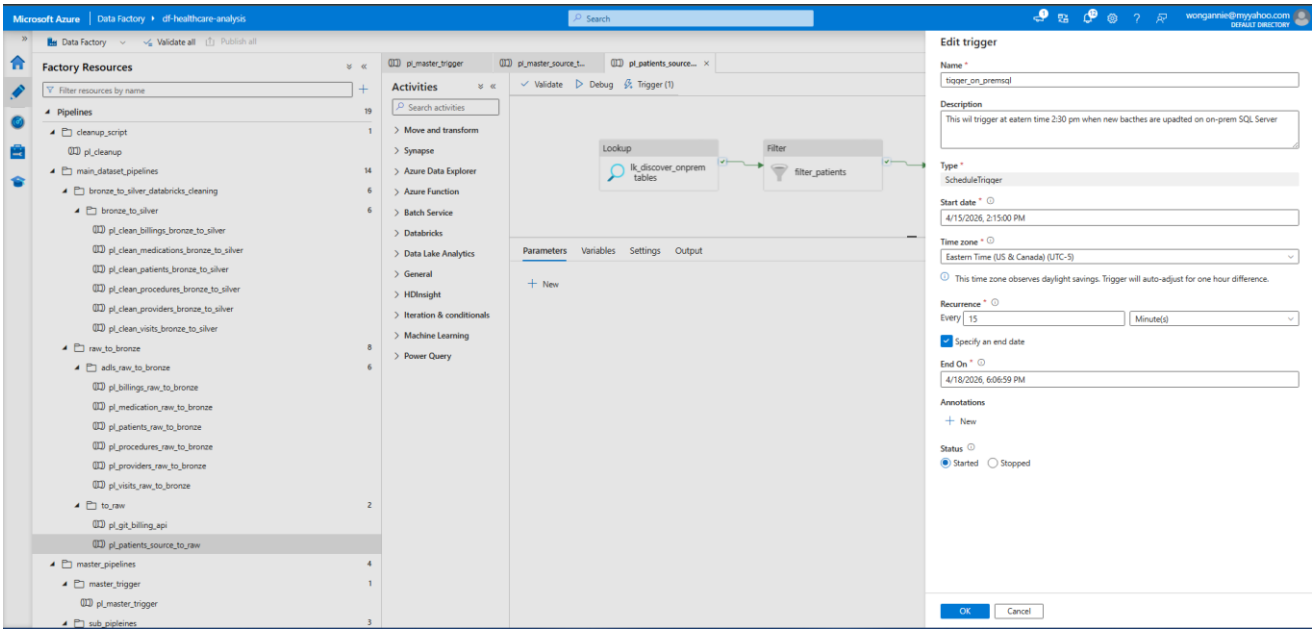


	Id	BatchName	TableName	Status	Layer	StartTime	EndTime
1	1	billing_batch1.json	Billing	Raw_Completed	Raw	2026-04-15T04:45:30.2970000	2026-04-15T04:46:51.9300000
2	2	patients_batch1	Patients	InProgress	Raw	2026-04-15T05:42:37.8000000	2026-04-15T05:05:54.6170000
3	3	patients_batch2	Patients	InProgress	Raw	2026-04-15T05:42:49.0230000	2026-04-15T05:05:51.0400000
4	4	billing_batch1	Billings	Bronze_Completed	Bronze	2026-04-15T05:05:59.6800000	2026-04-15T05:07:07.8470000
5	5	medications_batch1.csv	Medications	Bronze_Completed	Bronze	2026-04-15T05:32:14.5670000	2026-04-15T05:32:53.2600000
6	6	medications_batch2.csv	Medications	Bronze_Completed	Bronze	2026-04-15T05:32:30.0900000	2026-04-15T05:33:15.7070000
7	7	procedures_batch2	Procedures	Bronze_Completed	Bronze	2026-04-15T05:34:44.2370000	2026-04-15T05:35:28.1930000
8	8	procedures_batch1	Procedures	Bronze_Completed	Bronze	2026-04-15T05:34:48.9330000	2026-04-15T05:35:24.5630000
9	9	providers_batch1	Providers	Bronze_Completed	Bronze	2026-04-15T05:36:49.5000000	2026-04-15T05:37:39.1530000
10	10	visits_batch1	Visits	Bronze_Completed	Bronze	2026-04-15T05:39:28.4330000	2026-04-15T05:40:04.6030000
11	11	billings_batch2	Billings	Success	Silver	2026-04-15T05:39:59.9800000	2026-04-15T05:51:11.7430000
12	12	medications_batch3	Medications	Failed	Silver	2026-04-15T05:51:37.5930000	2026-04-15T05:52:37.3630000
13	13	billings_batch3	Billings	Success	Silver	2026-04-15T15:32:19.9530000	2026-04-15T15:41:24.5800000
14	14	billings_batch4	Billings	Success	Silver	2026-04-15T15:53:01.5930000	2026-04-15T15:53:48.4300000
15	15	medications_batch4	Medications	Failed	Silver	2026-04-15T15:54:18.9830000	2026-04-15T15:55:27.8270000
16	16	medications_batch5	Medications	Failed	Silver	2026-04-15T15:56:21.9470000	2026-04-15T15:57:44.5170000
17	17	patients_batch3	Patients	Failed	Silver	2026-04-15T15:59:19.7700000	2026-04-15T16:00:29.9500000
18	18	billings_batch5	Billings	Success	Silver	2026-04-15T16:05:01.8270000	2026-04-15T16:08:40.8600000

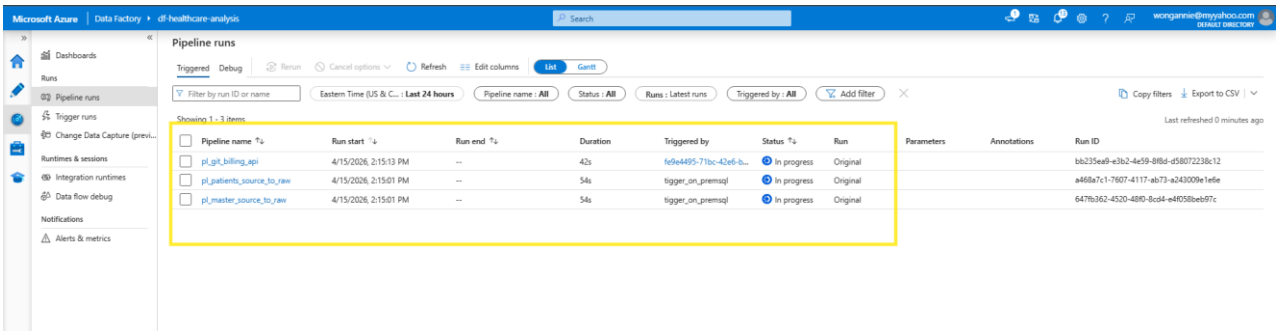
AuditLog table displaying batch-level status across Raw, Bronze, and Silver layers, highlighting successes and failures in pipeline execution.

TRIGGERED PIPLINE

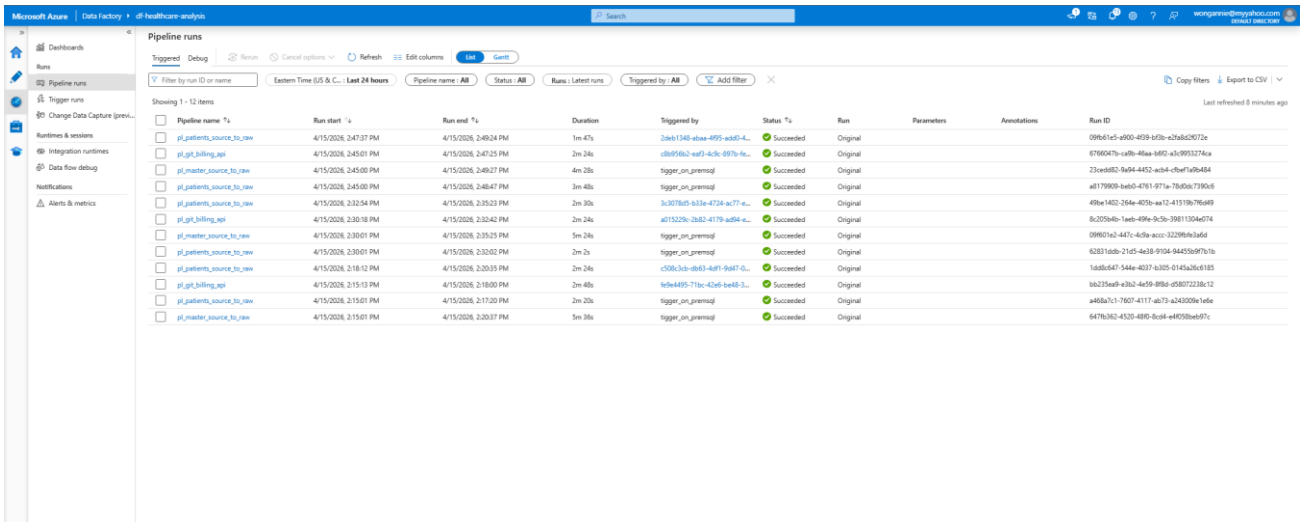
The pipeline was manually triggered at 2:15 PM to simulate a real-world data ingestion scenario where on-premises source systems are updated prior to scheduled processing. This controlled trigger helps validate the pipeline's ability to detect and process newly available data. Following this, the primary pipeline is scheduled to run automatically at 3:00 PM each day, ensuring a consistent and reliable ingestion pattern. During execution, data is ingested into the Bronze layer, preserving raw structure while standardizing formats for downstream processing. This approach ensures that the system can handle timing differences between data availability and pipeline execution, while maintaining data integrity, audit tracking, and readiness for further transformations in subsequent layers.



Configured ADF schedule trigger to execute the master pipeline at defined intervals, simulating timed ingestion from on-prem SQL sources and ensuring automated, consistent data flow into the Bronze layer for downstream processing.



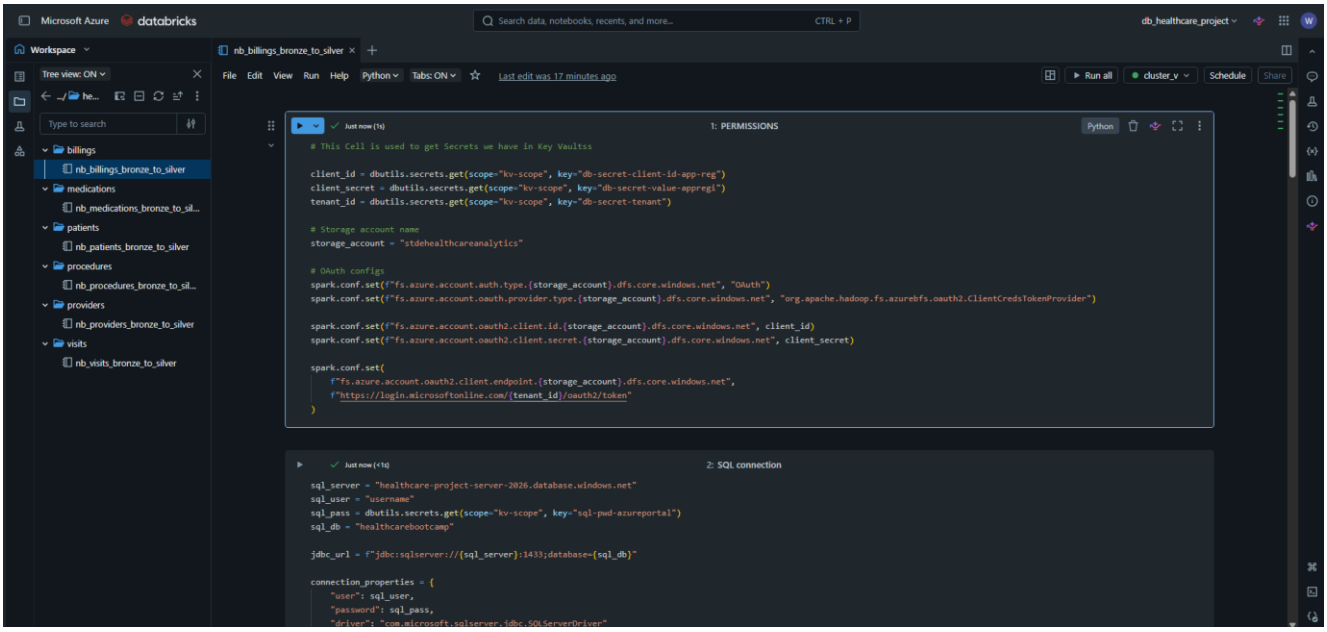
Pipeline runs successfully triggered via the scheduled trigger, showing multiple pipelines executing in parallel with “In Progress” status. This confirms that the automated scheduling is functioning as expected, initiating data ingestion workflows and ensuring coordinated execution of source-to-raw pipelines within the defined time window.



All triggered pipelines completed successfully with “Succeeded” status, confirming stable execution of scheduled ingestion workflows and reliable end-to-end data movement from source to raw layer without any failures.

DATABRICKS NOTEBOOKS

The Databricks notebook is used to perform data transformation and cleansing as part of the Bronze to Silver layer processing. It reads raw data from ADLS, applies necessary validations such as null handling, deduplication, and schema enforcement, and ensures data quality standards are met. The notebook also implements watermark logic using ingestion timestamps to process only new or updated records, making the pipeline idempotent and efficient. After transformation, the cleaned data is written to the Silver layer in Parquet format. This structured approach enables reliable downstream analytics, improves query performance, and ensures consistency across multiple pipeline runs.



Databricks notebook configuring secure access to ADLS using Azure Key Vault secrets and OAuth authentication, enabling protected data read/write operations through service principal credentials without exposing sensitive information.

```python
# SQL Server connection

sql_server = "healthcare-project-server-2026.database.windows.net"
sql_user = "username"
sql_pass = dbutils.secrets.get(scope="kv-scope", key="sql-pwd-azureportal")
sql_db = "healthcarebootcamp"

jdbc_url = f"jdbc:sqlserver://{sql_server}:1433;database={sql_db}"

connection_properties = {
    "user": sql_user,
    "password": sql_pass,
    "driver": "com.microsoft.sqlserver.jdbc.SQLServerDriver"
}
```

Databricks notebook establishing a secure connection to Azure SQL Server using JDBC, with credentials retrieved from Azure Key Vault, enabling safe and controlled access to relational data for downstream processing.

```python
from pyspark.sql.functions import col

# Read FilesTable
files_df = spark.read.jdbc(
    url=jdbc_url,
    table="dbo.FilesTable",
    properties=connection_properties
)

# Filter procedure files
pending_files_df = files_df.filter(
    (col("Status") == "Bronze_Processed") &
    (col("FileName").like("procedures%"))
)

# Create batch list
batch_list = [row.FileName for row in pending_files_df.select("FileName").collect()]

print("Files to process:", batch_list)

# Read from Bronze
bronze_path = "abfss://bronze@stdehealthcareanalytics.dfs.core.windows.net/procedures"

df = spark.read.format("parquet").load(bronze_path)

display(df.limit(10))
```

```
(3) Spark Jobs
df: pyspark.sql.dataframe.DataFrame = [procedure_id: string, patient_id: string ... 5 more fields]
files_df: pyspark.sql.dataframe.DataFrame = [Id: integer, FileName: string ... 4 more fields]
pending_files_df: pyspark.sql.dataframe.DataFrame = [Id: integer, FileName: string ... 4 more fields]
Files to process: []
```

Databricks notebook reading pending procedure files from the FilesTable using JDBC, filtering records based on Bronze_Processed status, and loading corresponding data from the Bronze layer in ADLS for further transformation and processing.

03:31 PM (1s) 4: Cleaning

```python
from pyspark.sql.functions import col, trim, when, regexp_replace

# Trim all string columns
for column in df.columns:
    df = df.withColumn(column, trim(col(column)))

# Empty strings to NULL
for column in df.columns:
    df = df.withColumn(column, when(col(column) == "", None).otherwise(col(column)))

# Remove unwanted characters
for column in df.columns:
    df = df.withColumn(
        column,
        when(col(column).isNotNull(),
            regexp_replace(col(column).cast("string"), "Â|\t", "")
        ).otherwise(col(column))
    )

# Replace "NULL" with actual NULL
df = df.replace("NULL", None)

display(df.limit(10))
```

(1) Spark Jobs

df: pyspark.sql.dataframe.DataFrame = [procedure_id: string, patient_id: string ... 5 more fields]

Table

	procedure_id	patient_id	visit_id	procedure_name	procedure_date	procedure_cost	procedure_outcome
1	PRO-17-0000001	PAT-15-0001521	VIS-15-0017193	Appendectomy	2017-06-03	4666.15	Inconclusive
2	PRO-17-0000002	PAT-15-0002929	VIS-15-0096797	Blood Panel	2018-12-05	5088.65	Follow-up Required
3	PRO-17-0000003	PAT-15-0001549	null	Physiotherapy	2018-11-07	716.39	Inconclusive
4	PRO-17-0000004	PAT-15-0003711	VIS-15-0049936	Blood Panel	null	3455.11	null
5	null	PAT-15-0001031	null	Blood Panel	2019-07-28	5715.71	Follow-up Required
6	PRO-17-0000006	PAT-15-0003727	VIS-15-0090619	Physiotherapy	2017-12-19	7696.87	Inconclusive
7	PRO-17-0000007	PAT-15-0004080	VIS-15-0057737	X-Ray Chest	2018-12-04	null	Follow-up Required
8	PRO-17-0000008	PAT-15-0000080	VIS-15-0082868	NaN	2018-05-23	327.79	null
9	PRO-17-0000009	PAT-15-0002069	VIS-15-0025528	NaN	2018-08-22	null	null
10	PRO-17-0000010	PAT-15-0000666	VIS-15-0018105	Appendectomy	2019-08-05	3948.59	Follow-up Required

10 rows | 0.78s runtime Refreshed 4 hours ago

Databricks notebook performing data cleansing by trimming whitespace, converting empty strings to NULL, removing unwanted characters, and standardizing values to ensure consistent, high-quality data before loading into the Silver layer.

Just now (1s) 6: READ FINAL SILVER

```python
silver_path = "abfss://silver@stdehealthcareanalytics.dfs.core.windows.net/procedures/silver"
spark.read.format("delta").load(f"{silver_path}/procedures_silver").display()
```

(1) Spark Jobs

Table

	procedure_id	patient_id	visit_id	procedure_name	procedure_date	procedure_cost	procedure_outcome
81	PRO-20-0015829	PAT-20-0002388	VIS-20-0142266	Physiotherapy	2021-07-06	3842.21	Follow-up Required
82	PRO-20-0015895	PAT-20-0003252	VIS-20-0199112	Appendectomy	2022-02-04	922.76	Success
83	PRO-20-0015938	PAT-20-0004070	null	null	2021-12-23	6605.99	Follow-up Required
84	PRO-20-0016168	PAT-20-0005811	VIS-20-0163165	Appendectomy	2022-12-05	NaN	Success
85	PRO-20-0016359	PAT-20-0006277	VIS-20-0060155	Blood Panel	2020-05-01	7213.89	Follow-up Required
86	PRO-20-0016825	PAT-20-0005063	VIS-20-0082113	Appendectomy	2022-02-19	3081.1	Follow-up Required
87	PRO-20-0017000	PAT-20-0005294	VIS-20-0105872	MRI Scan	2020-02-01	847.07	Success
88	PRO-20-0017227	PAT-20-0001056	VIS-20-0069067	Appendectomy	2020-02-22	5998.0	Follow-up Required
89	PRO-20-0018184	PAT-20-0006363	VIS-20-0245850	Blood Panel	2020-04-16	1533.7	Inconclusive
90	PRO-20-0018208	PAT-20-0005864	VIS-20-0119178	MRI Scan	2021-10-24	4071.12	Follow-up Required
91	PRO-20-0018378	PAT-20-0001224	VIS-20-0218705	null	2021-11-20	6683.22	null
92	PRO-20-0018629	null	VIS-20-0115066	Blood Panel	2022-10-09	6615.67	Follow-up Required
93	PRO-20-0018699	null	VIS-20-0162224	X-Ray Chest	2020-10-26	7031.79	Success
94	PRO-20-0018783	PAT-20-0000208	VIS-20-0238637	Appendectomy	2022-03-20	7535.97	Follow-up Required
95	PRO-20-0019873	PAT-20-0003871	VIS-20-0012083	MRI Scan	2022-04-14	4403.96	Follow-up Required

10,000+ rows | Truncated data | 0.69s runtime Refreshed now

Final Silver layer data successfully read from ADLS in Delta format, displaying cleaned, structured, and ready-to-use records after transformation, ensuring high-quality data is available for downstream analytics and reporting.

1 minute ago (1s) 7: Bad Records Table

```python
from pyspark.sql.functions import col

# Identify bad records
bad_df = df.filter(
    col("procedure_id").isNull() |
    col("procedure_name").isNull()
)

# Path
bad_path = "abfss://silver@stdehealthcareanalytics.dfs.core.windows.net/procedures/badrecords"

# Write bad records
bad_df.write.format("delta").mode("append").save(bad_path)

display(bad_df)
```

(2) Spark Jobs

bad_df: pyspark.sql.dataframe.DataFrame = [procedure_id: string, patient_id: string ... 5 more fields]

Table

	procedure_id	patient_id	visit_id	procedure_name	procedure_date	procedure_cost	procedure_outcome
1	null	PAT-20-0002439	VIS-20-0233791	Appendectomy	2020-02-18	1994.11	Inconclusive
2	PRO-20-0000007	PAT-20-0006642	null	null	null	3484.26	Follow-up Required
3	null	PAT-20-0002002	VIS-20-0154419	Physiotherapy	2021-11-06	NaN	Success
4	PRO-20-0000023	PAT-20-0000098	VIS-20-0150842	null	2021-06-24	2068.8	Success
5	PRO-20-0000026	null	VIS-20-0245697	null	null	6712.36	Inconclusive
6	PRO-20-0000028	PAT-20-0002717	null	null	2022-02-06	5587.36	Follow-up Required
7	PRO-20-0000042	PAT-20-0001517	VIS-20-0205006	null	null	2349.16	Inconclusive
8	PRO-20-0000043	PAT-20-0002323	VIS-20-0021091	null	2021-04-15	NaN	Success
9	PRO-20-0000045	PAT-20-0005440	VIS-20-0173168	null	2020-09-11	NaN	null
10	null	PAT-20-0002281	VIS-20-0007145	X-Ray Chest	2022-08-09	2248.64	Inconclusive
11	PRO-20-0000063	null	VIS-20-0079871	null	null	2365.85	Inconclusive
12	null	PAT-20-0005451	VIS-20-0119705	null	2021-09-06	2685.2	Follow-up Required
13	null	PAT-20-0004229	VIS-20-0056746	X-Ray Chest	2021-08-16	1949.6	Success
14	PRO-20-0000084	PAT-20-0005523	VIS-20-0216046	null	2021-11-01	NaN	Success
15	null	PAT-20-0000738	VIS-20-0131878	null	2022-02-28	7567.76	Inconclusive

10,000+ rows | Truncated data | 1.27s runtime Refreshed 2 minutes ago

Bad records identified based on missing critical fields (procedure_id or procedure_name) and stored separately in the Silver layer using Delta format, ensuring data quality by preventing invalid records from entering downstream processing.

CLEANED SILVER FILES

MESSY DATA

Just now (1s) 7: verification

```
# Check Silver
spark.read.format("delta").load(
    "abfss://silver@stdehealthcareanalytics.dfs.core.windows.net/provider/provider_silver"
).display()
```

(1) Spark Jobs

Table +

	provider_id	provider_name	department	specialization	experience_years	hospital_branch	provider_rating
1	NaN	Dr. Archer, Joshua	Oncology	Oncology	25	Kristenstad General Hospital	4.8
2	PRV-15-0000015	Dr. Hernandez, Jason	Orthopedics	Oncology	35	Reyeston General Hospital	3.2
3	PRV-15-0000030	Dr. Nguyen, Cassandra	Neurology	ER	12	New Davidside General Hospital	3.1
4	PRV-15-0000043	Dr. Young, Marissa	Oncology	Cardiology	29	Gibsonfurt General Hospital	2.1
5	PRV-15-0000050	Dr. Campbell, Tammy	Cardiology	null	23	New Kenneth General Hospital	1.8
6	PRV-15-0000091	Dr. Perez, Joshua	Cardiology	ER	23	Riveraburgh General Hospital	3.7
7	PRV-15-0000103	Dr. Wilson, David	Orthopedics	Orthopedics	29	Johnsonland General Hospital	null
8	PRV-15-0000119	Dr. Delgado, Christina	Orthopedics	Orthopedics	38	Lorraineland General Hospital	1.6
9	PRV-15-0000120	Dr. Thomas, Kimberly	null	null	null	Dianeville General Hospital	3.1
10	PRV-15-0000138	Dr. Lopez, Kenneth	Cardiology	Neurology	null	West Zachary General Hospital	3.4
11	PRV-15-0000178	Dr. Gonzales, Kevin	ER	Cardiology	null	Victorberg General Hospital	2.3
12	PRV-15-0000188	Dr. Wade, Angelica	Neurology	ER	26	Timothychester General Hospital	1.7
13	PRV-15-0000198	Dr. Mckinney, Julie	Neurology	ER	5	NaN	2.9
14	PRV-15-0000205	Dr. Hickman, Claudia	Pediatrics	Neurology	null	Lake Matthew General Hospital	null
15	PRV-15-0000220	Dr. Raymond, Christoph...	Pediatrics	Neurology	3	Sherrishire General Hospital	2.1

4,014 rows | 0.64s runtime Refreshed now

CLEANED DATA

Table +

	provider_id	provider_name	department	specialization	experience_years	hospital_branch	provider_rating
43	PRV-15-0000729	Dr. Burton, Seth	Orthopedics	Cardiology	30	Port Craigmouth General Hospital	4.2
44	PRV-15-0000734	null	Pediatrics	Neurology	31	Port Devinfort General Hospital	1.8
45	PRV-15-0000744	Dr. Olson, Justin	Neurology	Pediatrics	33	Coopershire General Hospital	4.2
46	PRV-15-0000749	Dr. Jones, Bryan	Neurology	Cardiology	39	Bruceton General Hospital	1.6
47	PRV-15-0000750	Dr. White, Kimberly	Cardiology	Oncology	23	Williamstown General Hospital	2.4
48	PRV-15-0000756	Dr. Rocha, Anthony	Neurology	Cardiology	15	Mcclainberg General Hospital	1.6
49	PRV-15-0000777	Dr. Taylor, Pamela	ER	Oncology	21	Port Raymondhaven General Hospital	3.2
50	PRV-15-0000778	Dr. Martinez, Jacob	Orthopedics	Pediatrics	32	Hessburgh General Hospital	null
51	PRV-15-0000779	Dr. Farmer, Larry	ER	Cardiology	13	New Timothyberg General Hospital	3.6
52	PRV-15-0000780	Dr. Gomez, Ricky	Cardiology	Pediatrics	24	NaN	2.8
53	PRV-15-0000790	Dr. Stewart, Gabriel	Oncology	null	17	Thomasfort General Hospital	1.4
54	PRV-15-0000808	Dr. Dean, Daisy	Orthopedics	Cardiology	null	NaN	4.7
55	PRV-15-0000809	Dr. Gonzales, Adam	Oncology	ER	9	East Joseph General Hospital	null
56	PRV-15-0000816	null	Orthopedics	Orthopedics	8	NaN	4.7
57	PRV-15-0000821	Dr. Smith, Robin	Orthopedics	Neurology	26	Christopherport General Hospital	4.4
58	PRV-15-0000844	Dr. Beasley, Christina	Pediatrics	ER	22	Jillhaven General Hospital	2.3
59	PRV-15-0000845	Dr. Best, Tammy	Orthopedics	Pediatrics	8	Christinechester General Hospital	null
60	PRV-15-0000852	Dr. Garcia, John	Oncology	Orthopedics	1	Morganview General Hospital	4.3
61	PRV-15-0000869	Dr. Bryant, Norma	Orthopedics	Oncology	null	NaN	2.2

Final validated Silver layer view displaying cleaned and standardized provider data after transformation, confirming successful data quality checks and readiness for downstream analytics and reporting.